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**UNIVERSITÀ
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Data Visualization Proposal

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Technical Report: City Bike Sharing System

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1 City Bike Dataset

Our main source of data is the City Bike dataset, which contains information about bike trips in New York City. These datasets are published on a monthly basis and can be accessed through the Citi Bike website¹.

The process involved in writing this report can be found in the Jupyter notebook in our repository².

1.1 Dataset Structure

The data is organized in multiple CSV files, each containing information about bike trips that ended in that month. Each row in the dataset represents a single bike trip and includes various attributes that describe the trip.

1.1.1 Original Structure

The original dataset includes the following columns:

- **Ride ID:** A unique identifier for each bike trip.
- **Rideable Type:** The type of bike used for the trip. Can be either "classic bike", "electric bike".
- **Started At:** a timestamp (YYYY-MM-DD HH:MM:SS) indicating when the trip started.
- **Ended At:** a timestamp indicating when the trip ended.
- **Start Info:** Information about the station and location where the trip started, structured as:
 - **Station Name:** The name of the station where the trip started.
 - **Station ID:** A unique identifier for the station where the trip started.
 - **Latitude:** The latitude coordinate of where the trip started.
 - **Longitude:** The longitude coordinate of where the trip started.
- **End Info:** Information about the station and location where the trip ended, structured similarly to the Start Info.

¹<https://www.citibikenyc.com/system-data>

²<https://github.com/446f6e6e79/nyc-bike-data-visualization/blob/main/src/notebooks/analysis.ipynb>

- **User Type:** The type of user who took the trip. Can be either "member" (a user with a subscription) or "casual" (a user without a subscription).

1.1.2 Extracted Features

These features alone are not sufficient for our analysis, so we extracted additional features from the original dataset, such as:

- **Trip Duration:** The duration of the trip in seconds, calculated as Ended At – Started At.
- **Trip Distance:** The distance traveled during the trip, computed from the latitude and longitude of the start and end stations using the Haversine formula. Various distance metrics were also tested, such as Manhattan distance, but the Haversine formula, multiplied by a circuitry factor of 1.3, provided the best results.
- **Day of the Week:** The day of the week on which the trip occurred, extracted from the "Started At" timestamp.
- **Hour of the Day:** The hour of the day when the trip started, also extracted from the "Started At" timestamp.
- **Station Popularity:** A measure of how frequently a station is used, calculated as the total number of trips that start or end at that station in a given time period (e.g hourly, daily).
- **Trips Flow:** The number of trips between pairs of stations, calculated by counting the number of trips that start at one station and end at another station in a given time period.

1.2 Live Data

1.3 Analysis and Insights

1.4 Data Quality

Various data quality issues were encountered during the analysis of the City Bike dataset.

Different Structure During the analysis, we noticed that older datasets have a different structure compared to newer ones. In particular, older dataset miss some of the columns that are present in newer datasets, such as the "Rideable Type", column in the "Start Info" and "End Info" fields. On the other hand, they include some columns that are not present in newer datasets, such as the Gender, Birth Year and Bike ID columns.

This inconsistency in the structure of the datasets posed a challenge for our analysis, as we had to adapt our data processing pipeline to handle both types of datasets and ensure that the extracted features were consistent across all datasets.

Missing Values and Inconsistencies

2 Weather Dataset

Our second source of data is hourly NYC weather, fetched live from the Open-Meteo archive API³.

2.1 Dataset Structure

The weather dataset captures hourly atmospheric conditions across New York City and serves as a proxy for environmental factors that influence cycling demand. Each record represents a one-hour observation aligned to Eastern Time (America/New_York), ensuring consistency with trip timestamps across the analysis.

2.1.1 Original Structure

The dataset from the Open-Meteo API includes the following core fields:

- **Timestamp:** the hourly observation time, localized to America/New_York timezone.
- **Temperature (temperature_2m):** ambient air temperature at 2 metres above ground level, measured in degrees Celsius.
- **Precipitation:** the cumulative precipitation during the hour, measured in millimetres. Most hours report zero; nonzero values are sparse.
- **Weather Code:** the WMO weather code, a standardized numeric classification that encodes condition types (clear, cloudy, rain, snow, etc.).
- **Wind Speed (wind_speed_10m):** instantaneous wind speed at 10 metres above ground level, measured in kilometres per hour.
- **Year:** the year of the observation, used for partitioning the dataset into annual Parquet files.

Unlike the City Bike dataset, where extracted features such as trip duration categories are derived during preprocessing, the weather dataset is

³<https://open-meteo.com/>

used largely in its raw form. The year column is the only structural addition, used for file partitioning rather than analysis. If additional discretized weather categories emerge during exploration (e.g., temperature bins for “cold,” “comfortable,” “hot”), they will be noted in the sections below.

Data are fetched from the Open-Meteo “archive” endpoint for the calendar year covering the ride data. The API request uses the NYC coordinates (`lat: 40.7823234`, `lon: -73.9654161`), requests hourly `fields temperature_2m, precipitation, weather_code` and `wind_speed_10m`, and sets `timezone` to `America/New_York` and `wind_speed_unit kmh`. The returned hourly time series is parsed into a dataframe with a datetime column and inspected for completeness (hours expected vs present).

2.2 Analysis and Insights

The exploratory analysis of the weather dataset is organised around two core questions: how weather variables are distributed and seasonal over time, and how they relate to ridership. The first perspective summarises hourly temperature, precipitation, wind, and the WMO weather codes using distributions and seasonality plots. Temperature exhibits a clear annual cycle with daily and seasonal variation; precipitation is highly skewed with most hours reporting zero and occasional spikes that drive short-term drops in ridership. Wind shows moderate variation and can amplify the perceived discomfort at low temperatures.

The second perspective connects weather to demand. Simple aggregated plots relating average hourly ridership to binned temperature reveal the expected pattern: ridership increases with temperature up to a comfortable range and declines in cold conditions. Precipitation hours show a consistent reduction in ridership, with heavier hourly totals producing larger drops. Weather codes can be used as categorical annotations (clear, rain, snow) to summarise combined effects: rainy or snowy hours have lower ridership even at similar temperatures.

Operationally, hourly ridership is computed by flooring trip start times to the hour and counting rides per hour. This rides-per-hour series is inner-joined to the weather hourly dataframe on the matching datetime; temperature is grouped into 5°C bins (e.g. 0–4°C, 5–9°C) to compute mean rides per bin and visualise the non-linear relation between temperature and demand.

These analyses justify including the weather dataset: it provides explanatory variables that help account for systematic variation in cycling demand and improves the interpretability of temporal ridership patterns.

2.3 Data Quality

Overall the archived hourly feed is high quality: timestamps are complete and consistently aligned to America/New_York, and the core variables are recorded at regular hourly intervals. Important caveats remain. The feed effectively represents a single observation point (Central Park or the archive's chosen station), so it may not capture microclimate variation across the five boroughs. This spatial simplification can introduce bias when local weather differs from the station's measurements, but it is a pragmatic choice for citywide analysis.

Precipitation is sparse and right-skewed: many hours report zero and a few hours contain most of the cumulative rainfall. When plotting histograms or computing summary statistics, non-zero precipitation should be examined separately to avoid misleading central tendencies. Temperature and wind are continuous and well-behaved, but gusts or short anomalies can appear and should be checked against outlier thresholds if they drive unusual model behaviour.

Practical notes from the exploration: histograms show temperature and wind distributions across the year; precipitation plots focus on non-zero hours to avoid the mass at zero. Mean monthly temperature plots clearly show the annual cycle. When joining to rides the notebook confirms a rise in mean hourly ridership with temperature up to a comfortable band and a drop in cold hours; rainy or snowy weather corresponds to lower ridership at similar temperatures.

Finally, the year column is an engineered field used for partitioning Parquet files and does not affect analysis directly. If derived categorical features (e.g. temperature bins or precipitation intensity classes) are created they should be documented and justified in the analysis methods.

3 Bike Lanes Dataset

Our third source of data is NYC's bike-route network, published by NYC OpenData⁴ as dataset `mzxcg-pwib`. It is used to render bike infrastructure (lanes, paths, and their protection level) in the visualization.

3.1 Dataset Structure

The bike-lanes dataset describes New York City's bicycle infrastructure at the segment level. Unlike the trip dataset, where each row is an individual ride, here each row corresponds to a physical piece of the network: a lane, path, or related facility drawn as a line on the map. This makes the dataset

⁴<https://data.cityofnewyork.us/d/mzxcg-pwib>

especially useful for spatial analysis, because the raw table is already close to its visual form. The core fields tell us where a segment is, what kind of facility it is, whether it is still active, and when it was installed or retired.

From the perspective of the final visualization, this dataset provides the infrastructure layer. It is the source of the map geometry, but also of the attributes that control how each segment is classified and styled. A protected route segment should be interpreted differently from a shared lane, and a current segment should be treated differently from a retired one. For that reason, the dataset is not just a geographic background: it is an analytical object that lets us talk about the structure and evolution of the bike network itself.

3.1.1 Original Structure

The original CSV contains one row per route segment and includes the following relevant attributes:

- **Segment ID:** the identifier of the route segment within the source dataset.
- **Bike ID:** a bike-network identifier used by the source to track the same segment across revisions.
- **Status:** whether the segment is currently active or has been retired from service.
- **Installation Date:** the date on which the segment was added to the network.
- **Retirement Date:** the date on which the segment was removed or replaced, when available.
- **the_geom:** the segment geometry, stored as a line representation and used to draw the route on the map.
- **Street:** the street where the segment is located.
- **From Street:** the starting cross street or intersection reference.
- **To Street:** the ending cross street or intersection reference.
- **Facility Class:** the level of protection or treatment provided by the segment. This field is central to the visualization, because it distinguishes protected paths from lower- protection or shared facilities such as sharrows lanes.
- **Borough:** the borough in which the segment lies.

In the analysis, this structure is used to derive three families of summaries: counts by borough, counts by status, and counts by facility class. These summaries help reveal how the network is distributed across the city, which parts of the network are current, and which kinds of facilities are most common. They also provide the rationale for how the map layer should be read and styled.

The raw file also contains a few columns that are not necessary for the analysis presented in this report. The fields `prevbikeid`, `gwsys2`, `spur`, `ft2facilit`, and `tf2facilit` were dropped during preprocessing because they do not contribute to the network breakdowns or to the map rendering used in the final visualization.

3.2 Analysis and Insights

This dataset is the spatial counterpart to the trip data. If the trip dataset explains demand, the bike-lanes dataset explains supply: where cycling infrastructure exists, what level of protection it offers, and how the network has changed over time. In the report, the exploration is organised around three questions that directly connect to the final map.

The first question is where the network is concentrated. A borough-level breakdown shows that the largest number of segments are in Manhattan and Brooklyn, while the outer boroughs contain fewer segments. This does not merely describe a count; it reveals the geography of cycling investment. Dense segment counts in central boroughs imply a more mature and interconnected infrastructure, while lower counts elsewhere suggest a sparser network or a more fragmented build-out. The borough view therefore helps explain why the final map appears most visually dense in the core of the city.

The second question is what kind of facilities dominate the network. The facility-class distribution is important because it encodes the degree of separation between cyclists and traffic. Protected paths and protected lanes support a very different riding experience from shared lanes or sharrows, and that difference matters both analytically and visually. In the map, this field can be used to assign visual priority, making the safer or more substantial infrastructure stand out. In the analysis, it also helps us understand whether the city has expanded primarily through high-comfort infrastructure or through lighter interventions that share space with motor vehicles.

The third question is how the network has grown over time. The installation timeline shows the build-up of the system year by year and gives a sense of the pace of infrastructure expansion. Sudden increases in the number of installed segments usually correspond to citywide policy pushes, corridor projects, or larger protected-lane programmes. This timeline is useful for

interpretation because it contextualises the current network: a dense cluster of segments on the map may reflect decades of incremental growth, or it may be the result of a recent acceleration in capital investment.

The map of the network is the key output of the dataset. By plotting `the_geom`, the analysis turns the table into a geographic layer that can be overlaid with trip demand and station data. The map is not only descriptive but also diagnostic: it shows whether the dataset is spatially coherent, where segments cluster, and how the network spans the city. Segments concentrate visually in Manhattan and Brooklyn, reflecting the geographic distribution observed in the borough counts. In the final visualization, this layer becomes the infrastructure context against which trip flows and station activity can be interpreted.

Taken together, these analyses show that the bike-lanes dataset is useful for more than just drawing a line map. It supports a structural reading of the network, a historical reading of its expansion, and a cartographic reading of its spatial coverage. These three views are complementary and together justify why the dataset is included as a separate source in the project.

3.3 Data Quality

The bike-lanes data is comparatively clean for a public GIS-style dataset, but it still needs preprocessing and careful interpretation.

One data-management issue is column redundancy. Several fields in the raw CSV are not relevant to the analysis or to the final visualization. The removed columns `prevbikeid`, `gwsys2`, `spur`, `ft2facilit`, and `tf2facilit` do not change the core structure of the segment table, so dropping them makes the dataset easier to work with without reducing analytical value.

Status handling is another important quality consideration. The raw file includes both current and retired segments, and those records should not be mixed blindly. Retired segments are useful for understanding how the network evolved, but they should be excluded or clearly separated when the goal is to depict the active bike network. Otherwise, the map would overstate the amount of usable infrastructure.

Date fields are also uneven. Installation and retirement dates may be missing for some records, and this especially affects historical segments. As a result, the installation timeline should be read as a slightly conservative view of growth: years with missing dates are undercounted, and the earliest phases of the network may appear smaller than they really were.

The geometry field deserves special attention as well. Because `the_geom` stores the route as a line geometry, malformed or missing values prevent a segment from being rendered. Those rows are not useful for the map layer

and must be filtered or repaired before plotting. In practice, this is the main reason the analysis uses a sampled set of segments for the interactive map: rendering the full dataset at once would be heavier and less responsive, while a sample is sufficient to demonstrate the overall spatial pattern.

4 Station Metadata Dataset (GBFS)

Our fourth source of data is current station information and live availability from the Lyft General Bikeshare Feed Specification (GBFS) feed⁵, ie. the same feed used by the backend. Two endpoints are merged: `station_information` (static: name, location, capacity) and `station_status` (live: bikes/docks available, operational flags).

4.1 Dataset Structure

The station metadata feed combines two GBFS endpoints into a single table used by the backend. The static endpoint provides the descriptive properties of each station, while the live endpoint reports the current availability and operational state.

4.1.1 Original Structure

The merged dataset is organized as follows:

- **From `station_information` (static):**
 - **Station ID:** a numeric GBFS identifier unique to the feed.
 - **Short Name:** the public station identifier used in the Citi Bike trip data, which is the key needed for cross-dataset joins.
 - **Name:** the full station name.
 - **Latitude** and **Longitude:** the station coordinates.
 - **Capacity:** the total number of docks at the station.
- **From `station_status` (live):**
 - **Bikes available:** the current number of bikes that can be rented.
 - **Docks available:** the current number of empty docks.
 - **Operational flags:** `is_installed`, `is_renting`, and `is_returning`, which indicate whether the station is active and usable.

⁵<https://gbfs.lyft.com/gbfs/2.3/bkn/en/gbfs.json>

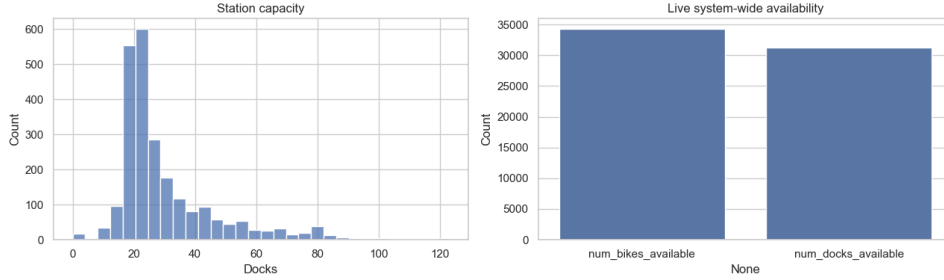


Figure 1: Distribution of station capacity (number of docks) and live availability (bikes and docks available).

4.1.2 Extracted Features

The backend also derives an **active** indicator from `is_installed`, `is_renting`, and `is_returning`. This feature is useful for filtering out inactive stations in live views and for avoiding stations that are temporarily unavailable.

4.2 Analysis and Insights

The exploratory analysis of the station metadata is organised around three questions: how station sizes are provisioned, how availability varies in real-time, and how stations are distributed geographically. The capacity distribution reveals whether the network relies on many small stations or a smaller number of large hubs; this informs aggregation choices and marker scaling in map views. The live availability metrics (bikes and docks available) reveal temporal imbalance and saturation: a high fraction of empty or full stations indicates a redistribution need and suggests using saturation-aware color scales or alerts in the dashboard. Finally, plotting coordinates on a city map exposes spatial clustering, gaps in coverage, and the relationship between station capacity and surrounding infrastructure.

Figure 1 summarises the numeric distributions used in the analysis : the histogram of station capacities, and paired distributions for bikes and docks available at query time. These plots are used to select sensible axis limits, bin widths, and outlier-handling rules for the final figures.

The spatial view is equally important. Mapping station points with size or color encoding for capacity or current availability reveals both macro-level patterns (e.g. densification in central boroughs) and micro-level issues (isolated large stations, small stations adjacent to each other). In the analysis this map is used iteratively to validate joins with trip data, identify coordinate errors, and decide on spatial binning strategies for aggregate views.

Beyond static summarisation, the analysis also inspects temporal snapshots and short time-series aggregated at the station level. These lightweight time-

Figure 2: Map of stations in New York City, colored by their active state.

series highlight rapid depletion or replenishment events and are useful for designing live-updating widgets or simple predictive indicators in the final dashboard.

4.3 Data Quality

The main data-quality issues found in practice fall into three categories: operational flags, missing attributes, and identifier consistency.

Operational flags: a portion of stations are reported in the feed but are effectively unusable because one or more of `is_installed`, `is_renting`, or `is_returning` are false. The analysis quantifies these cases and shows how an `active` boolean (derived from the three flags) reduces noise in live dashboards. For live monitoring the recommended approach is to filter by `active==true` and surface a count of inactive stations to the operator.

Missing attributes: a small number of records lack capacity or coordinate values. Missing capacity prevents inclusion in capacity histograms or in size encodings on the map, while missing coordinates render a station unusable for spatial overlays. The analysis documents the frequency of these gaps and applies conservative filters: stations without coordinates are excluded from map layers, and stations without capacity are retained only for non-capacity analyses or imputed when appropriate.

Identifier consistency: joins across datasets require care. The feed exposes a numeric `station_id` and a public-facing `short_name`. Historical trip data use the `short_name`, so all cross-dataset joins in the analysis are performed on `short_name` after validating uniqueness and normalising formatting. This mirrors the inconsistency discussed in the City Bike chapter and motivates adding a small mapping table in the backend to ensure stable joins over time.

Other considerations noted in the analysis include transient measurement error during feed outages (which appear as sudden drops in reported availability), clock skew across sources, and duplicate short names caused by legacy station renames. Each of these issues is handled through clear preprocessing steps and documented thresholds so that the visualization remains reliable.